

Introduction to Logic

Assignment 5

King Mongkut's Institute of Technology Ladkrabang

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Problem 1

Suppose $\Gamma = \{p \rightarrow (s \vee t), (p \wedge s) \rightarrow q, (t \wedge \neg q) \rightarrow \perp\}$

1.1 Show by means of a truth table that $\Gamma \models p \rightarrow q$.

1.2 Show by constructing a natural-deduction derivation that $\Gamma \vdash p \rightarrow q$.

Problem 2

Each passage below contains an argument. For each passage, please do the following:

- Write the underlined statements in the passage in propositional logic using the given propositional letters and its specified meaning.
- From the formulas you obtained in (a), determine which formulas are the premises and which formula is the conclusion of the argument in the passage.
- Based on what you identified as the premises and the conclusion in (b), determine whether the argument is valid or not. If so, provide a derivation of the conclusion from the premises using natural deduction rules. If not, give a truth assignment which makes all the premises true but the conclusion false.

Example. ¹John must not be at home at the moment. ²If he were at home, his car must be in the garage. But from what I can see, ³his car is currently not in the garage.

h = John is at home at the moment.

g = John's car is currently in the garage.

Ans.

- Statement 1 = $\neg h$
Statement 2 = $h \rightarrow g$
Statement 3 = $\neg g$
- Premises: $h \rightarrow g, \neg g$

Conclusion: $\neg h$

(c) The argument is valid.

1 : $h \rightarrow g$	premise
2 : $\neg g$	premise
3 : $\neg h$	<i>MT</i> , 1, 2

2.1 ¹If Virginia supports independence, then so do the southern colonies. ²If Virginia and the southern colonies support independence, then the northern colonies will also support independence. Therefore, ³the northern and the southern colonies will both support independence if Virginia does.

v = Virginia supports independence.

s = The southern colonies support independence.

n = The northern colonies support independence.

2.2 ¹If high-tech products are exported to Russia, then domestic industries will benefit. ²If the Russians can effectively utilize high-tech products, then their standard of living will improve. Therefore, ³if high-tech products are exported to Russia and the Russians can effectively utilize them, then their standard of living will improve and domestic industries will benefit.

h = High-tech products are exported to Russia.

d = Domestic industries will benefit.

u = The Russians can effectively utilize high-tech products.

s = The Russians' standard of living will improve.

2.3 ¹The deaths were caused either by overdoses of heroin or by bad quality heroin. ²If the former, the victims would have shown the usual overdose symptoms. However, ³they did not exhibit these symptoms. Therefore, we may conclude that ⁴the deaths were caused by bad quality heroin.

o = The deaths were caused by overdoses of heroin.

b = The deaths were caused by bad quality heroin.

s = The victims showed the usual overdose symptoms.

2.4 ¹Either Alex or David (or both) is a thief. ²If Alex is a thief, then Bob is also a thief. And ³if Bob is a thief, so is Calvin. ⁴If Alex and David are both thieves, then Calvin is also a thief. Therefore, ⁵if David is a thief, so is Calvin.

a = Alex is a thief.

b = Bob is a thief.

c = Calvin is a thief.

d = David is a thief.

2.5 ¹If Cain married his sister, his marriage was incestuous. ²If he did not marry his sister, then Adam and Eve were not the progenitors of the entire human race. It follows that ³if Adam and Eve were the progenitors of the whole human race, then Cain's marriage was incestuous.

s = Cain married his sister.
 i = Cain's marriage was incestuous.
 p = Adam and Eve were the progenitors of the entire human race.

2.6 ¹If Japan is to reduce its huge trade surplus, then it must either convince its citizens to spend more or it must move its manufacturing facilities to other countries. ²It is not the case that Japan will either increase its imports or convince its citizens to spend more. Furthermore, ³it is not the case that Japan will either allow foreign companies to compete fairly or move its manufacturing facilities to other countries. Therefore, ⁴Japan will not reduce its huge trade surplus.

s = Japan will reduce its huge trade surplus.
 i = Japan will increase its imports.
 c = Japan will convince its citizens to spend more.
 f = Japan will allow foreign companies to compete fairly.
 m = Japan will move its manufacturing facilities to other countries.

2.7 ¹You should not stay up all night to study for the exam. ²If you do that, you will be tired in the morning. And ³if you are tired in the morning and the exam is difficult, you will not be able to do well on the exam. Obviously, ⁴if you are not able to do well on the exam even though you stay up all night to study, then why do that?

s = You should stay up all night to study for the exam.
 u = You stay up all night to study for the exam.
 t = You are tired in the morning.
 d = The exam is difficult.
 w = You are able to do well on the exam.

2.8 ¹Watson has reddish dirt on his boots. ²He wouldn't have that if he had not been to the Post Office this morning. ³If he had been to the Post Office but did not mail a letter this morning, then either he bought some stamps or sent a telegram. ⁴If he mailed a letter this morning, then he would have written a letter this morning. ⁵He wouldn't buy some stamps unless he ran out of stamps. Therefore, ⁶if Watson didn't write a letter this morning and didn't run out of stamps, then he must have sent a telegram this morning.

d = Watson has reddish dirt on his boots.
 p = Watson went to the Post Office this morning.
 m = Watson mailed a letter this morning.
 s = Watson bought some stamps this morning.
 t = Watson sent a telegram this morning.
 w = Watson wrote a letter this morning.
 o = Watson ran out of stamps.

2.9 ¹If there is evil and God does not know it, then God is not omniscient. ²If there is evil and God knows it but he is unable to prevent it, then God is not omnipotent. ³If there is evil and God knows it and is able to prevent it but is unwilling to do so, then God is not supremely good. ⁴If God exists and there is evil, then either God does not know it or he is unable or unwilling to prevent it. ⁴If God exists then he is omnipotent, omniscient, and

supremely good. It follows that ⁵either there is no evil or there is no God (or both). (Yu Kam Por)

e = There is evil.

g = There is God.

k = God knows that there is evil.

a = God is able to prevent evil.

w = God is willing to prevent evil.

c = God is omniscient.

p = God is omnipotent.

s = God is supremely good.

Problem 1

Suppose $\Gamma = \{p \rightarrow (s \vee t), (p \wedge s) \rightarrow q, (t \wedge \neg q) \rightarrow \perp\}$

1.1 Show by means of a truth table that $\Gamma \models p \rightarrow q$.

1.2 Show by constructing a natural-deduction derivation that $\Gamma \vdash p \rightarrow q$.

1.1/

p	q	s	t	$\neg q$	$s \vee t$	premise $p \rightarrow (s \vee t)$	$p \wedge s$	premise $(p \wedge s) \rightarrow q$	$t \wedge \neg q$	premise $(t \wedge \neg q) \rightarrow \perp$	conclusion $p \rightarrow q$
T	T	T	T	F	T	T	T	T	F	T	T
T	T	T	F	F	T	T	T	T	F	T	T
T	T	F	T	F	T	T	F	T	F	T	T
T	T	F	F	F	F	F	F	T	F	T	T
T	F	T	T	T	T	T	T	F	T	F	F
T	F	T	F	T	T	T	T	F	F	T	F
T	F	F	T	T	T	T	F	T	T	F	F
T	F	F	F	T	F	F	F	T	F	T	F
F	T	T	T	F	T	T	F	T	F	T	T
F	T	T	F	F	T	T	F	T	F	T	T
F	T	F	T	F	T	T	F	T	F	T	T
F	T	F	F	F	F	T	F	T	F	T	T
F	F	T	T	T	T	T	F	T	T	F	T
F	F	T	F	T	T	T	F	T	F	T	T
F	F	F	T	T	T	T	F	T	T	F	T
F	F	F	F	T	F	T	F	T	F	T	T

$\therefore \Gamma \models p \rightarrow q$

1.2/

1. $p \rightarrow (s \vee t)$ premise
2. $(p \wedge s) \rightarrow q$ premise
3. $(t \wedge \neg q) \rightarrow \perp$ premise

4.	p	Ass
5.	$s \vee t$	MP 1, 4
6.	s	Ass
7.	$p \wedge s$	$\wedge I$ 4, 6
8.	q	MP 2, 7
9.		
10.	t	Ass
	$\neg q$	Ass
	$t \wedge \neg q$	$\wedge I$ 6, 7
	$\perp$	MP 3, 8
	q	PBC 7-9

$\therefore \Gamma \vdash p \rightarrow q$

11. q VE 5, 6-10
12. $p \rightarrow q$ $\rightarrow I$ 4-11

2.1 ¹If Virginia supports independence, then so do the southern colonies. ²If Virginia and the southern colonies support independence, then the northern colonies will also support independence. Therefore, ³the northern and the southern colonies will both support independence if Virginia does.

$v \rightarrow (n \wedge s)$
 v = Virginia supports independence.
 s = The southern colonies support independence.
 n = The northern colonies support independence.

(a) Statement 1 = $v \rightarrow s$

Statement 2 = $(v \wedge s) \rightarrow n$

Statement 3 = $v \rightarrow (s \wedge n)$

(b) Premises: $v \rightarrow s$, $(v \wedge s) \rightarrow n$

Conclusion: $v \rightarrow (s \wedge n)$

(c)

1.	$v \rightarrow s$	Premise
2.	$(v \wedge s) \rightarrow n$	Premise
3.	v	Ass
4.	s	MP 1,3
5.	$v \wedge s$	$\wedge I$ 3,4
6.	n	MP 2,5
7.	$s \wedge n$	$\wedge I$ 4,6
8.	$v \rightarrow (s \wedge n)$	$\rightarrow I$ 3-7

$\therefore$ The argument is valid.

2.2 ¹If high-tech products are exported to Russia, then domestic industries will benefit. ²If the Russians can effectively utilize high-tech products, then their standard of living will improve. Therefore, ³if high-tech products are exported to Russia and the Russians can effectively utilize them, then their standard of living will improve and domestic industries will benefit.

$h \rightarrow d$
 $u \rightarrow s$
 $h \wedge u \rightarrow (s \wedge d)$
 h = High-tech products are exported to Russia.
 d = Domestic industries will benefit.
 u = The Russians can effectively utilize high-tech products.
 s = The Russians' standard of living will improve.

(a) Statement 1 = $h \rightarrow d$

Statement 2 = $u \rightarrow s$

Statement 3 = $(h \wedge u) \rightarrow (s \wedge d)$

(b) Premises: $h \rightarrow d$, $u \rightarrow s$

Conclusion: $(h \wedge u) \rightarrow (s \wedge d)$

(c)

1.	$h \rightarrow d$	Premise
2.	$u \rightarrow s$	Premise
3.	$h \wedge u$	Ass
4.	h	$\wedge E$ 3
5.	d	MP 1,4
6.	u	$\wedge E$ 3
7.	s	MP 2,6
8.	$s \wedge d$	$\wedge I$ 5,7
9.	$(h \wedge u) \rightarrow (s \wedge d)$	$\rightarrow I$ 3-8

$\therefore$ The argument is valid.

2.3 ¹The deaths were caused either by overdoses of heroin or by bad quality heroin. ²If the former, the victims would have shown the usual overdose symptoms. However, ³they did not exhibit these symptoms. Therefore, we may conclude that ⁴the deaths were caused by bad quality heroin.

- o = The deaths were caused by overdoses of heroin.
 b = The deaths were caused by bad quality heroin.
 s = The victims showed the usual overdose symptoms.

(a) Statement 1 = $o \vee b$

Statement 2 = $o \rightarrow s$

Statement 3 = $\neg s$

Statement 4 = b

(c)

- | | | |
|----|-------------------|---------|
| 1. | $o \vee b$ | Premise |
| 2. | $o \rightarrow s$ | Premise |
| 3. | $\neg s$ | Premise |
| 4. | $\neg o$ | MT 2,3 |
| 5. | b | DS 1,4 |

$\therefore$ The argument is valid.

- | | | |
|----|-------------------|--------------|
| 1. | $o \vee b$ | Premise |
| 2. | $o \rightarrow s$ | Premise |
| 3. | $\neg s$ | Premise |
| 4. | o | Ass |
| 5. | s | MP 2,4 |
| 6. | $\perp$ | $\neg E$ 3,5 |
| 7. | $\neg o$ | $\neg I$ 4-6 |
| 8. | b | DS 1,7 |

(b) Premises: $o \vee b, o \rightarrow s, \neg s$

Conclusion: b

2.4 ¹Either Alex or David (or both) is a thief. ²If Alex is a thief, then Bob is also a thief. And ³if Bob is a thief, so is Calvin. ⁴If Alex and David are both thieves, then Calvin is also a thief. Therefore, ⁵if David is a thief, so is Calvin.

- a = Alex is a thief.
 b = Bob is a thief.
 c = Calvin is a thief.
 d = David is a thief.

(a) Statement 1 = $a \vee d$

Statement 2 = $a \rightarrow b$

Statement 3 = $b \rightarrow c$

Statement 4 = $(a \wedge d) \rightarrow c$

Statement 5 = $d \rightarrow c$

- (c)
- | | | |
|-----|------------------------------|------------|
| T1. | $a \vee d$ | Premise |
| T2. | $a \rightarrow b$ | Premise |
| T3. | $b \rightarrow c$ | Premise |
| T4. | $(a \wedge d) \rightarrow c$ | Premise |
| 5. | $d \rightarrow c$ | Conclusion |

(b) Premises: $a \vee d, a \rightarrow b, b \rightarrow c, (a \wedge d) \rightarrow c$

Conclusion: $d \rightarrow c$

				Premises				Conclusion	
a	b	c	d	$a \vee d$	$a \rightarrow b$	$b \rightarrow c$	$(a \wedge d) \rightarrow c$	$d \rightarrow c$	
F	F	F	T	T	T	T	T	F	$\therefore$ The argument is not valid.

2.5 ^{s → i} If Cain married his sister, his marriage was incestuous. ^{¬s} If he did not marry his sister, then Adam and Eve were ^{¬i} not the progenitors of the entire human race. It follows that ^{¬s} if Adam and Eve were the progenitors of the whole human race, then Cain's marriage was incestuous.

s = Cain married his sister.
 i = Cain's marriage was incestuous.
 p = Adam and Eve were the progenitors of the entire human race.

(a) Statement 1 = $s \rightarrow i$
 Statement 2 = $\neg s \rightarrow \neg p$
 Statement 3 = $p \rightarrow i$

(b) Premises: $s \rightarrow i, \neg s \rightarrow \neg p$
 Conclusion: $p \rightarrow i$

(c)	1. $s \rightarrow i$	Premise
	2. $\neg s \rightarrow \neg p$	Premise
	3. p	Ass
	4. $\neg \neg p$	$\neg \neg I$ 3
	5. $\neg \neg s$	MT 2, 4
	6. s	$\neg \neg E$ 5
	7. i	MP 1, 6
	8. $p \rightarrow i$	$\rightarrow I$ 3-7

$\therefore$ The argument is valid.

2.6 ^{s →} If Japan is to reduce its huge trade surplus, then it must either convince ^{c ∨ m} its citizens to spend more or it must move its manufacturing facilities to other countries. ^{¬s} It is not the case that Japan will either increase its imports or convince its citizens to spend more. Furthermore, ^{¬(f ∨ m)} it is not the case that Japan will either allow foreign companies to compete fairly or move its manufacturing facilities to other countries. Therefore, ^{¬s} Japan will not reduce its huge trade surplus.

s = Japan will reduce its huge trade surplus.
 i = Japan will increase its imports.
 c = Japan will convince its citizens to spend more.
 f = Japan will allow foreign companies to compete fairly.
 m = Japan will move its manufacturing facilities to other countries.

(a) Statement 1 = $s \rightarrow (c \vee m)$
 Statement 2 = $\neg(i \vee c)$
 Statement 3 = $\neg(f \vee m)$
 Statement 4 = $\neg s$

(b) Premises: $s \rightarrow (c \vee m), \neg(i \vee c), \neg(f \vee m)$
 Conclusion: $\neg s$

(c)	1. $s \rightarrow (c \vee m)$	Premise
	2. $\neg(i \vee c)$	Premise
	3. $\neg(f \vee m)$	Premise
	4. s	Ass
	5. $c \vee m$	MP 1, 4
	6. c	Ass
	7. $i \vee c$	VI 6
	8. $\perp$	$\perp E$ 2, 7
	9. $\perp$	VE 5-8
	10. $\neg s$	$\neg I$ 4-9

$\therefore$ The argument is valid.

2.7 ¹You should not stay up all night to study for the exam, ²If you do that, you will be tired in the morning. And ³if you are tired in the morning and the exam is difficult, you will not be able to do well on the exam. Obviously, ⁴if you are not able to do well on the exam even though you stay up all night to study, then why do that? $\neg w \wedge u \rightarrow \neg s$

s = You should stay up all night to study for the exam.

u = You stay up all night to study for the exam.

t = You are tired in the morning.

d = The exam is difficult.

w = You are able to do well on the exam.

(a) Statement 1 = $\neg s$

Statement 2 = $u \rightarrow t$

Statement 3 = $(t \wedge d) \rightarrow \neg w$

Statement 4 = $(\neg w \wedge u) \rightarrow \neg s$

(b) Premises: $u \rightarrow t, (t \wedge d) \rightarrow \neg w, (\neg w \wedge u) \rightarrow \neg s$

Conclusion: $\neg s$

(c)

$\begin{matrix} T & u \rightarrow t \\ T & (t \wedge d) \rightarrow \neg w \\ T & (\neg w \wedge u) \rightarrow \neg s \\ F & \neg s \end{matrix}$

The argument is not valid because the exam may not be difficult. In case the exam is not difficult, it is possible that we can do the exam even though we are tired in the morning.

					Premises			Conclusion
s	t	u	w	d	$u \rightarrow t$	$(t \wedge d) \rightarrow \neg w$	$(\neg w \wedge u) \rightarrow \neg s$	$\neg s$
T	T	T	T	F	T	T	T	F

$\therefore$ The argument is not valid.

2.8 ¹Watson has reddish dirt on his boots. ²He wouldn't have that if he had not been to the Post Office this morning. ³If he had been to the Post Office but did not mail a letter this morning, then either he bought some stamps or sent a telegram. ⁴If he mailed a letter this morning, then he would have written a letter this morning. ⁵He wouldn't buy some stamps unless he ran out of stamps. Therefore, ⁶if Watson didn't write a letter this morning and didn't run out of stamps, then he must have sent a telegram this morning. (¬w ∧ ¬o) → t

d = Watson has reddish dirt on his boots.
 p = Watson went to the Post Office this morning.
 m = Watson mailed a letter this morning.
 s = Watson bought some stamps this morning.
 t = Watson sent a telegram this morning.
 w = Watson wrote a letter this morning.
 o = Watson ran out of stamps.

(a) Statement 1 = d

Statement 2 = $\neg p \rightarrow \neg d$

Statement 3 = $(p \wedge \neg m) \rightarrow (s \vee t)$

Statement 4 = $m \rightarrow w$

Statement 5 = $\neg o \rightarrow \neg s$

Statement 6 = $(\neg w \wedge \neg o) \rightarrow t$

Premises: $d, \neg p \rightarrow \neg d, (p \wedge \neg m) \rightarrow (s \vee t), m \rightarrow w, \neg o \rightarrow \neg s$

Conclusion: $(\neg w \wedge \neg o) \rightarrow t$

(c)

1. d	}	Premises
2. $\neg p \rightarrow \neg d$		
3. $(p \wedge \neg m) \rightarrow (s \vee t)$		
4. $m \rightarrow w$		
5. $\neg o \rightarrow \neg s$		

6. $\neg w \wedge \neg o$	Ass
7. $\neg w$	$\wedge E$ 6
8. $\neg m$	MT 4, 7
9. $\neg o$	$\wedge E$ 6
10. $\neg s$	MP 5, 9
11. $\neg \neg d$	$\neg \neg I$ 1
12. $\neg \neg p$	MT 2, 11
13. p	$\neg \neg E$ 12
14. $p \wedge \neg m$	$\wedge I$ 8, 13
15. $s \vee t$	MP 3, 14
16. t	DS 10, 15

17. $(\neg w \wedge \neg o) \rightarrow t$ $\rightarrow I$ 6-16

$\therefore$ The argument is valid.

2.9 ¹If there is evil and God does not know it, then God is not omniscient. ²If there is evil and God knows it but he is unable to prevent it, then God is not omnipotent. ³If there is evil and God knows it and is able to prevent it but is unwilling to do so, then God is not supremely good. ⁴If God exists and there is evil, then either God does not know it or he is unable or unwilling to prevent it. ⁵If God exists then he is omnipotent, omniscient, and supremely good. It follows that ⁵either there is no evil or there is no God (or both). (Yu Kam Por)

$\neg e \vee \neg g$

- e = There is evil.
- g = There is God.
- k = God knows that there is evil.
- a = God is able to prevent evil.
- w = God is willing to prevent evil.
- c = God is omniscient.
- p = God is omnipotent.
- s = God is supremely good.

(a) Statement 1 = $(e \wedge \neg k) \rightarrow \neg c$

Statement 2 = $(e \wedge k \wedge \neg a) \rightarrow \neg p$

Statement 3 = $(e \wedge k \wedge a \wedge \neg w) \rightarrow \neg s$

Statement 4 = $(g \wedge e) \rightarrow (\neg k \vee \neg a \vee \neg w)$

Statement 5 = $g \rightarrow (p \wedge c \wedge s)$

Statement 6 = $\neg e \vee \neg g$

(b) Premises: $(e \wedge \neg k) \rightarrow \neg c, (e \wedge k \wedge \neg a) \rightarrow \neg p, (e \wedge k \wedge a \wedge \neg w) \rightarrow \neg s,$

$(g \wedge e) \rightarrow (\neg k \vee \neg a \vee \neg w), g \rightarrow p \wedge c \wedge s$

Conclusion: $\neg e \vee \neg g$

$e \wedge k \wedge \neg a \rightarrow \neg p$
 $e \wedge k \wedge a \wedge \neg w \rightarrow \neg s$
 $g \wedge e \rightarrow (\neg k \vee \neg a \vee \neg w)$
 $g \rightarrow p \wedge c \wedge s$

1	$e \wedge \neg k \rightarrow \neg c$				<i>given</i>
2	$e \wedge k \wedge \neg a \rightarrow \neg p$				<i>given</i>
3	$e \wedge k \wedge a \wedge \neg w \rightarrow \neg s$				<i>given</i>
4	$g \wedge e \rightarrow \neg k \vee \neg a \vee \neg w$				<i>given</i>
5	$g \rightarrow p \wedge c \wedge s$				<i>given</i>
6	$g \vee \neg g$				<i>LEM</i>
7	g	$< ass >$	$\neg g$	$< ass >$	
8	$p \wedge c \wedge s$	$\rightarrow E(5, 7)$	$\neg e \vee \neg g$	$\vee I(7)$	
9	p	$\wedge E(8)$			
10	c	$\wedge E(8)$			
11	s	$\wedge E(8)$			
12	e	$< ass >$			
13	$\neg k$	$< ass >$			
14	$e \wedge \neg k$	$\wedge I(12, 13)$			
15	$\neg c$	$\rightarrow E(1, 14)$			
16	$\perp$	$\perp I(10, 15)$			
17	k	$PBC(13 - 16)$			
18	$\neg a$	$< ass >$			
19	$e \wedge k$	$\wedge I(12, 17)$			
20	$e \wedge k \wedge \neg a$	$\wedge I(18, 19)$			
21	$\neg p$	$\rightarrow E(2, 20)$			
22	$\perp$	$\perp I(9, 21)$			
23	a	$PBC(18 - 22)$			
24	$g \wedge e$	$\wedge I(7, 12)$			
25	$\neg k \vee \neg a \vee \neg w$	$\rightarrow E(4, 24)$			
26	$\neg k$	$< ass >$	$\neg a$	$< ass >$	$\neg w$
27	$\perp$	$\perp I(17, 26)$	$\perp$	$\perp I(23, 26)$	$e \wedge k$
28					$e \wedge k \wedge a$
29					$e \wedge k \wedge a \wedge \neg w$
30					$\neg s$
31					$\perp$
32	$\perp$	$\vee E(25, 26 - 31)$			
33	$\neg e$	$\neg I(12 - 32)$			
34	$\neg e \vee \neg g$	$\vee I(33)$			
35	$\neg e \vee \neg g$				$\vee E(6, 7 - 34)$